

TAPER TANTRUM 2013

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What was 'Taper Tantrum' that hit financial markets in 2013?

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Taper Tantrum

[ˈtā-pər ˈtān-trəm]

The 2013 surge in U.S. Treasury yields, resulting from the Federal Reserve's (Fed) announcement of future tapering of its policy of quantitative easing.



- Taper tantrum was an episode in 2013 that started with Fed Chair, Ben Bernanke hinting at future tapering of its QE bond buying programme. US bond yields surged immediately and there were large spillovers to other countries, especially emerging markets which witnessed capital outflows.
- Starting with bond sell-off, the EM exchange rates faced depreciation pressures and then mayhem spread to equity markets as portfolio investors chose to sell-off equities as interest rate stocks were hit and then weakening EM currencies eroded foreign investor's profits. Derivative markets, especially futures, provided empathies to bears in the financial markets.
- It was not a full-scale financial crisis but a near one that could have turned into a prolonged one but for corrective central bank communications. There was a latent fear that the market would collapse if QE is withdrawn.
- Brazil, India, Indonesia, South Africa, Turkey became the fragile five that were worst impacted by market turmoil

FOMC dilemma in the year of ‘Taper Tantrum’: to taper or Not

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- In the 29-30th Jan, 2013 FOMC meeting, policymakers debated the benefits of third round of QE that had started in Sept 2012. Powell called for ending QE by year-end by tapering bond purchases. While FOMC voted to continue QE, Esther George of Fed Kansas casted first of her 7 dissent votes of 2013.
- George again dissented at 19-20th March & 30th April-1st May meetings, while Powell pushed for taper to start in June while again not dissenting.
- **On 22nd May, 2013, Bernanke in his Congressional testimony stated that “If we see continued improvement and we have confidence that that’s going to be sustained then we could in the next few meetings ... take a step down in our pace of purchases,”** **Taper tantrum** began in the markets; EM yields soared, equities & currencies plunged. Bond-currency-equities-derivatives nexus operated.
- 18-19th June FOMC statement avoided clear communication on taper amid jittery markets though George dissented again but Ben Bernanke used post FOMC press conference to reiterate taper possibility by emphasizing that any changes to bond buys would depend on the economic outlook, and that decisions about bond purchases were distinct from those concerning interest rates. Specifically, he said that the Fed could scale back its bond purchases from \$85 billion to \$65 billion a month during the upcoming September 2013 policy meeting. He also suggested that the bond-buying program could wrap up by mid-2014. While Bernanke did not announce an interest rate hike, he suggested that if inflation followed a 2% target rate and unemployment decreased to 6.5%, the Fed would likely start raising rates (Evans Rule).
- The stock markets dropped by ~4.3% over the three trading days following Bernanke's announcement, with the Dow Jones dropping 659 points between 19 and 24 June, closing at 14,660 at the end of the day on 24 June.

FOMC compelled to dither Taper amid market turmoil

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Emerging Market Currencies vs. U.S. Dollar (Index 5/22/2013 = 100)

Brazil
Indonesia
Turkey
South Africa

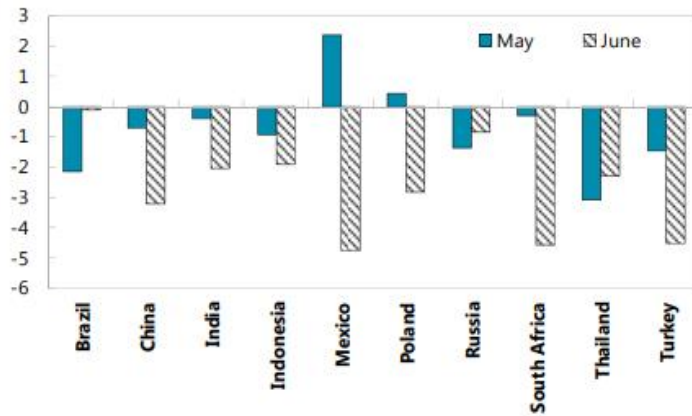


- In his 17-18th July Congressional testimony, Bernanke explained that Fed remained committed to accommodative monetary policy even after it slowed bond purchases and it could keep interest rates near zero even after unemployment fell below 6.5% (tanking Evans Rule)
- 30-31st July & 17-18th Sept FOMC meetings, maintained QE and held back taper with George dissenting and Powell advocating in Sept that it was time to taper
- On 16th Oct, in an unscheduled VC meeting Fed discussed how it may respond in case of US debt default in case Congress did not lift the debt ceiling, but Congress voted to raise debt ceiling later that day.
- On 29-30th Oct, Esther George dissented for the 7th time after FOMC again delayed taper
- On 17-18th Dec, 2013 FOMC announced it would begin to taper its purchases in January 2014. it would reduce its monthly pace of asset purchases to \$75 billion, from \$85 billion, and added that it would continue to reduce the bond purchases in “further measured steps” if the economy continued to perform as expected. George voted for the policy, but Boston Fed President Eric Rosengren dissented, calling the move premature.
- Market volatility limited during actual taper as markets had priced it by then. Tapering completed with purchases were halted on 29th Oct 2014

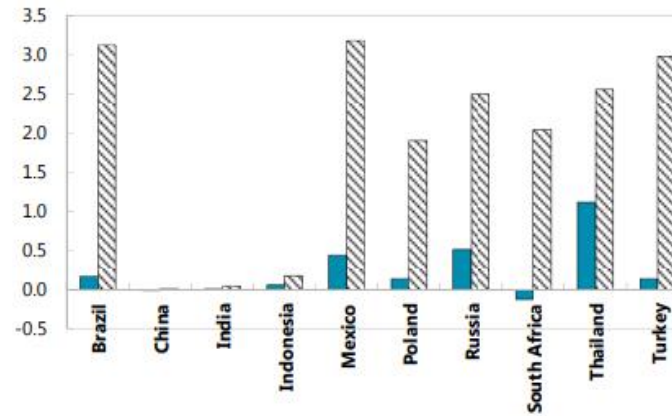
The mayhem in EM markets following the Fed's taper indication is captured by Z-scores for variations in asset prices due to May-June 2013 monetary policy shock

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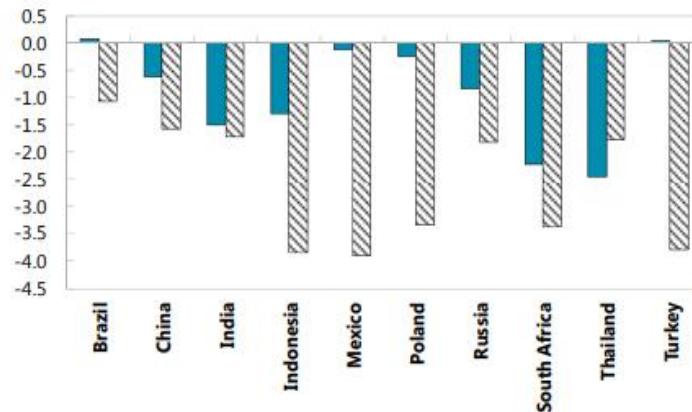
Capital Flows
(Bond & Equity)



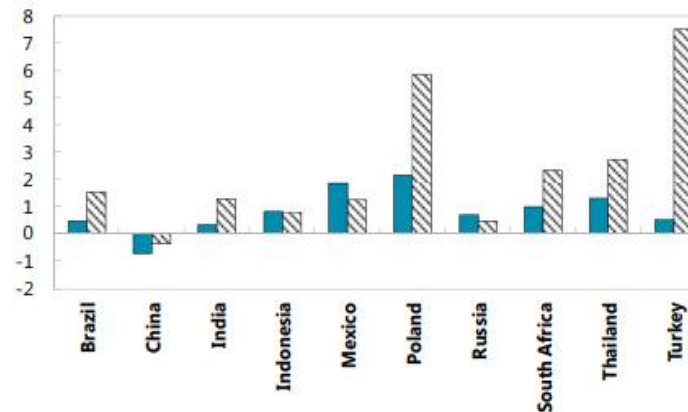
Exchange Rate



Equity Price



10-year Bond Yield



- $z = (x - \mu) / \sigma$, where 'x' is the data point, ' μ ' is the mean, and ' σ ' is the S.D.
- Z-score indicates how many standard deviations a data point is away from the mean of a dataset. It can be positive or negative, indicating whether the data point is above or below the mean, respectively. Z-scores are useful for comparing values from different datasets and for determining the probability of a value occurring within a normal distribution
- A z-score of 0 indicates that the data point is equal to the mean. A positive z-score indicates that the data point is above the mean. A negative z-score indicates that the data point is below the mean. A z-score of 1 means the data point is one standard deviation above the mean. A z-score of -2 means the data point is two standard deviations below the mean

How did 'Taper Tantrum' hit India during 2013?

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India was amongst fragile five essentially because of its macroeconomic vulnerabilities.

- **Growth had moderated from 8.5% in 2010-11 to 5.2% in 2011-12 and 5.5% in 2012-13** before was seen improving marginally in 2013-14 on back of good agriculture, electricity and 'financing, insurance, real estate & business services' (GDP growth turned out to be 6.4% in 2013-14)
- **CPI inflation rose, peaking at 11.2% in Nov. 2013.** Apart from the pressure from rising food prices, the sharp depreciation of the rupee in the H1 of 2013-14 led to pass-through effects, especially in the case of freely priced fuel products. Staggered revisions in administered prices of diesel also added to inflationary impulses. Even more worryingly, CPI ex-food and fuel remained sticky at around 8% for most of 2013-14. Double digit inflation in housing and 'others' sub-category. Inflation moderated subsequently only after seasonal food-price corrections but was still at 8.% in March 2014.
- Twin deficit concerns were looming large. **Fiscal slippages looked apparent in H1 of 2013-14 with projections suggesting that centre's GFD may cross 6.0% of GDP from just under 5.0% in the previous year.** However, expenditure restraints, improved tax and non-tax revenues due to improved activity in H2 eventually helped restraint GFD/GDP at 4.5% in 2013-14.
- **CAD had widened alarmingly in Q1 of 2013-14 at 4.9% of GDP**, a correction in CAD. Subsequently helped compress full-year CAD to 1.7% of GDP. This transformation after two years of wide CAD was brought about by policy interventions, especially curbs on gold imports and softening of international commodity prices such as oil, fertilisers and coal. Slow growth also kept imports in check.

External Vulnerability Indicators Had deteriorated

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		(per cent)			
Indicator		End-Mar 2011	End-Mar 2012	End-Mar 2013	End-Mar 2014
1		2	3	4	5
1.	External Debt to GDP	18.2	20.5	22.0	23.3
2.	Ratio of Short-term to Total Debt (Original Maturity)	20.4	21.7	23.6	20.3
3.	Ratio of Short-term to Total Debt (Residual maturity)#	40.6	40.9	42.1	39.6
4.	Ratio of Concessional Debt to Total Debt	14.9	13.3	11.1	10.5
5.	Ratio of Reserves to Total Debt	95.9	81.6	71.3	69.0
6.	Ratio of Short-term Debt to Reserves	21.3	26.6	33.1	29.3
7.	Ratio of Short-term Debt (Residual maturity) to Reserves#	42.3	50.1	59.0	57.4
8.	Reserves Cover of Imports (in months)	9.5	7.1	7.0	7.8
9.	Debt Service Ratio (Debt Service Payments to Current Receipts)	4.4	6.0	5.9	5.9
10.	External Debt (US\$ billion)	317.9	360.8	409.4	440.6
11.	International Investment Position (IIP) (US\$ billion)	-207.0	-264.7	-326.7	-331.6
12.	IIP/GDP ratio	-11.9	-14.8	-17.6	-17.6
13.	CAD/GDP ratio	2.8	4.2	4.7	1.7
#: RBI estimate.					

Raghuram Rajan's first press conference as 23rd RBI Governor proved to be a turning point

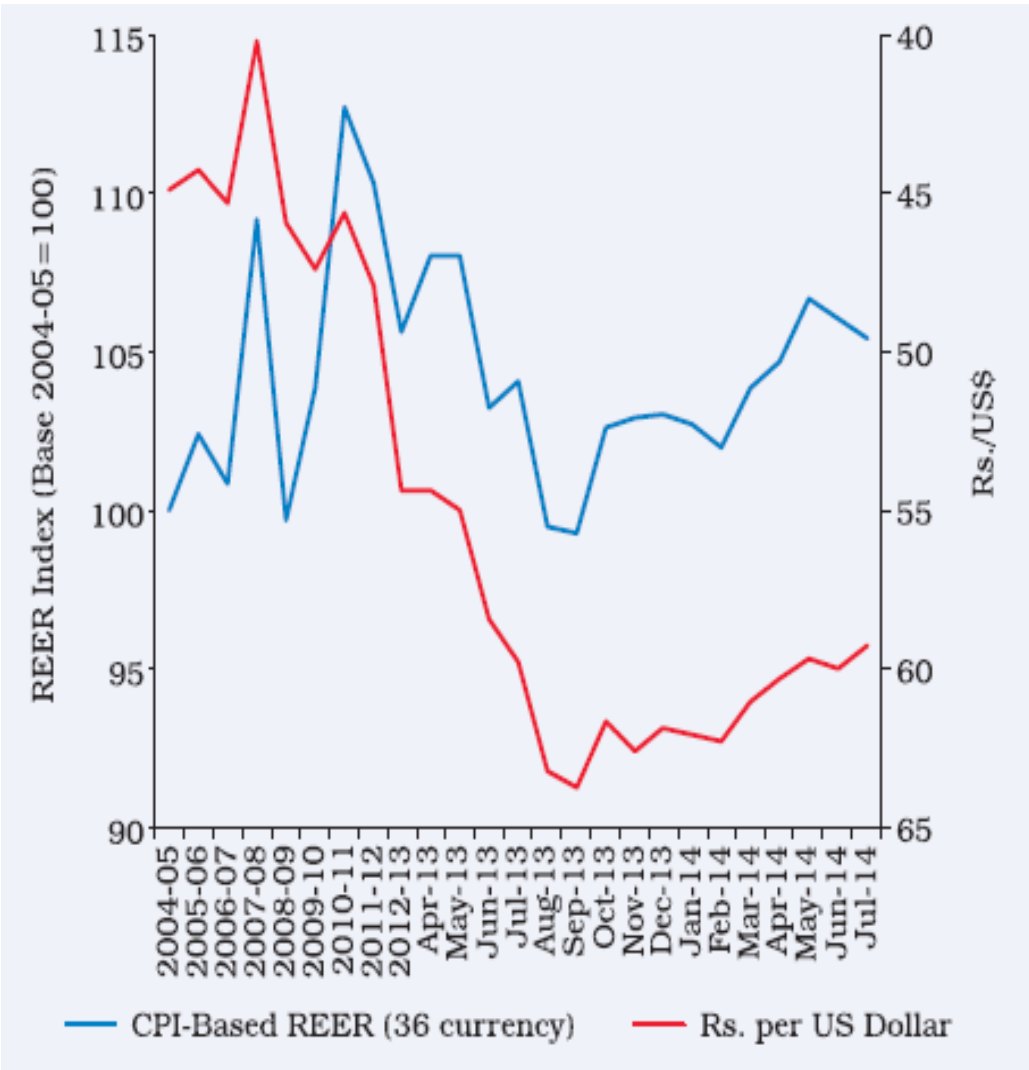
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- On 6th August 2013, Raghuram Rajan was appointed as the new RBI Governor with a 3-year term. He stated that “we don’t have a magic wand to make the problems disappear instantaneously. But I have absolutely no doubt, we will deal with them”
- Raghuram Rajan had impeccable credentials. He was the Katherine Dusak Miller Distinguished Service Professor of Finance at the University of Chicago’s Booth School of Business. Between 2003 and 2006 he was Economic Counsellor and Director of the Research Department at IMF, a prestigious position commonly known as the Chief Economist of the IMF.
- In 2005, ahead of GFC unfolding, Rajan at the Fed Kansas Jackson Hole conference, warned about the growing risks in the financial system, that a financial crisis could be in the offing, and proposed policies that would reduce such risks.
- On 3rd September 2013 he took charge as 23rd RBI Governor taking over from Druvvi Subbarao at a time when the Indian markets were badly hit by taper tantrum.

<https://www.youtube.com/watch?v=nW0ZhN6kqIA>

How badly were the Indian markets hit in 2013? INR & IDR amongst the worst performing EM currencies in a year to end-June 2013



Exchange Rates against the US dollar			
	(Per cent)		
	End-Aug 2013 over end-Jun 2013	End-Dec 2013* over end-Aug 2013	Aug 12, 2014 over end-Dec 2013*
1	2	3	4
Argentine Peso	-5.0	-13.1	-21.1
Brazilian Real	-7.3	0.2	3.3
Chinese Yuan	0.1	1.1	-0.8
Euro	1.2	4.1	-3.2
Indian Rupee	-10.3	7.4	1.3
Indonesian Rupiah	-9.1	-10.6	4.5
Malaysian Ringgit	-3.7	0.2	3.1
Mexican Peso	-2.4	2.1	-0.6
Russian Rouble	-1.6	1.6	-9.3
South African Rand	-3.1	-1.6	-1.6
South Korean Won	3.5	5.2	2.4
Thai Baht	-3.0	-2.3	2.3
Turkish Lira	-5.3	-4.1	-1.6

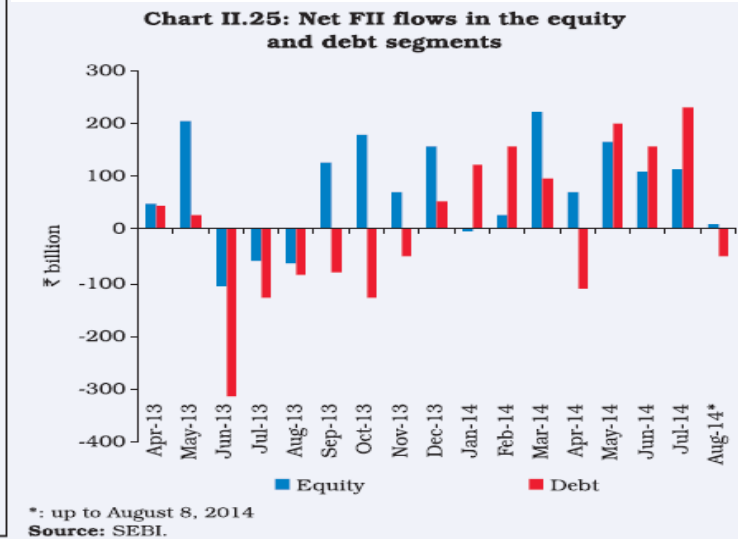
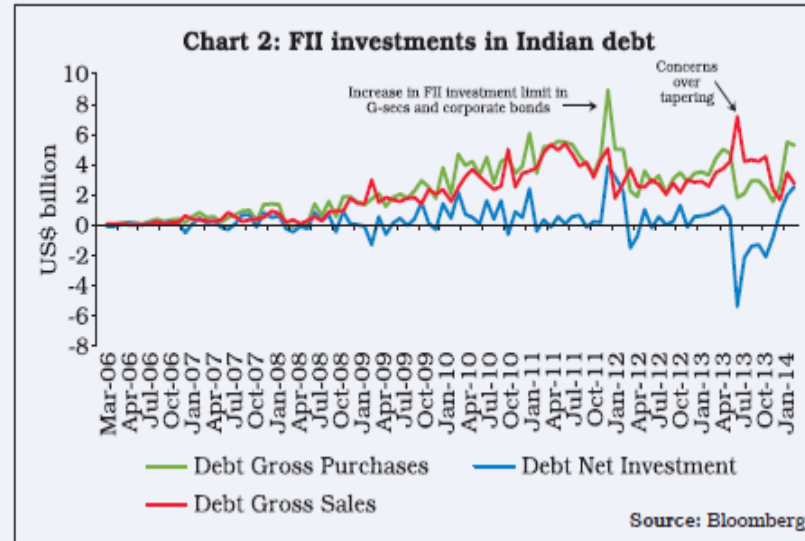
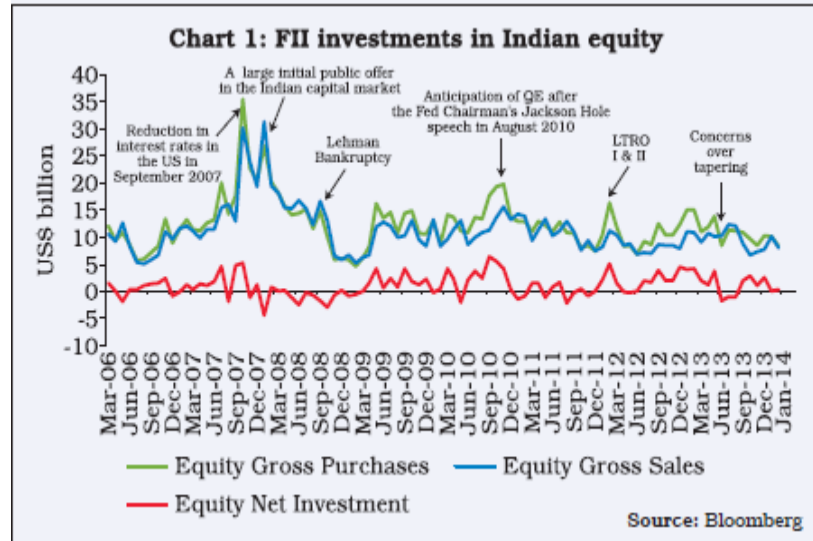
*: Data for end-December 2013 pertain to December 30, 2013.

INR/USD had depreciated 17.9% between May 21 to Sept 4 of 2013 but appreciated 8.3% in the rest of the year. Currency crisis was prevented by FX reserves depletion of US\$17 bn till Sept 4.

How badly were the Indian markets hit in 2013?

FII sell offs, portfolio outflows continued through June-Nov. 2013

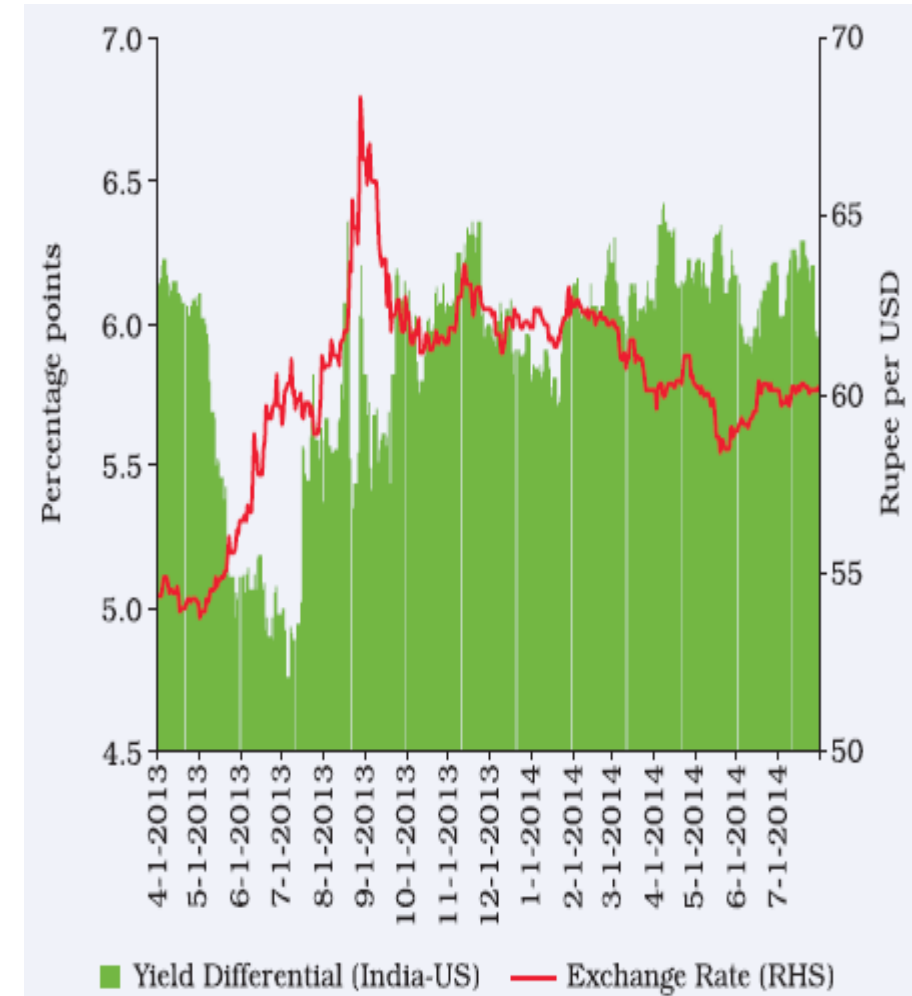
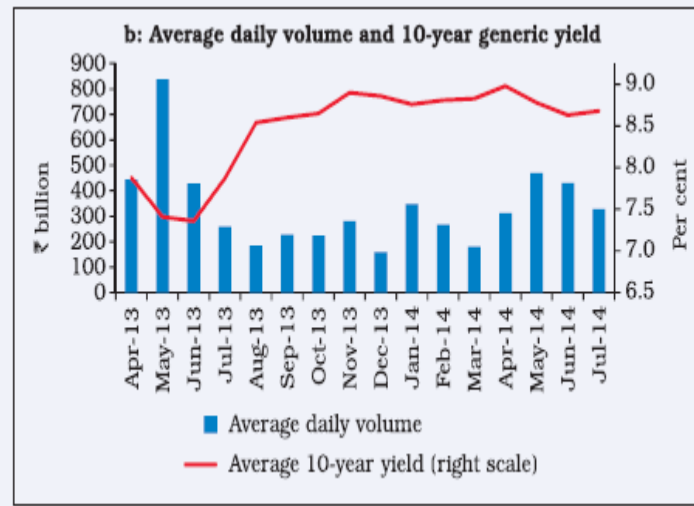
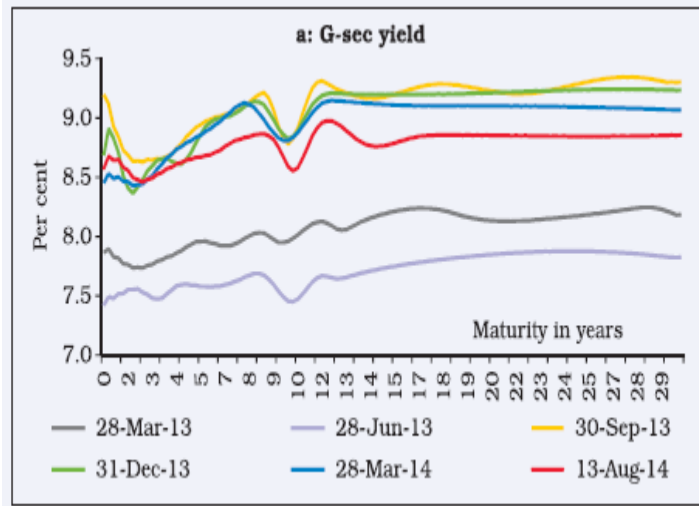
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- There were net FII disinvestments of over US\$13.4 bn (US\$10.5 bn in debt and US\$2.8 bn in equity) since the testimony till early Sept 2013
- The equity market remained subdued during May-Aug 2013 but started recovering following a number of reform measures announced since Sept. 2013.
- During H2 of 2013-14, the BSE Sensex and the NSE Nifty recorded gains of 20.2% and 22.5%, respectively on expectations of formation of a stable government after the Lok Sabha elections in May 2014. For the year 2013, SENSEX recorded 9% gain on top of 26% gain in the preceding year.

How badly were the Indian markets hit in 2013? Bond sell-offs were sizeable during May-July 2013 as interest rate differentials narrowed

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- G-sec yields rose during 2013-14, while the yield curve remained flat with inversions at few maturities. The Fed's taper talk in May/June 2013 altered global monetary and financial conditions dramatically with spillovers in the Indian markets bringing to the fore the challenge of conducting monetary policy in a globalised world. With unexpectedly significant hardening of long-term yields in the US, the yield differential (between 10-year G-sec yields in India and the US) fell from more than 6.2 percentage points (ppt) before May 2013 to less than 5 ppt in June and July 2013. The 10-year generic yield peaked at 9.27% on August 19, 2013.

Key policy actions taken by India during spillovers from Taper Talk Tantrum

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- Series of **liquidity tightening measures** taken (open market operation (OMO) sales of G-secs of Rs. 120 bn was announced in July 2013; **LAF limited to 1.0% of NDTL at a system level and later to 0.5% of NDTL for each bank**; banks were asked to **maintain daily 99% of the average fortnightly requirement**).
- Marginal Standing Facility (MSF) rate at the top of the interest rate corridor was hiked 200 bps to 10.25% from 8.25%. Effectively this meant that **operational policy rate moved from 7.25% repo rate to 10.25% MSF rate – a 300 bps rate stealth tightening**. Interest rate defence of currency became the policy choice, but was not very effective due to dominance of equity in portfolio flows
- **Series of CFMs** were also taken with steep hike in import duty on gold imports to eventually 10%, and 80:20 scheme for gold imports under which import of gold by nominated agencies required at least 20% of import of gold was needed to be exclusively made available for the purpose of export.
- **RBI FCNR(B) swap window opened** in which fresh deposits of minimum 3-year tenor were swapped at a fixed rate of 3.5% p.a., reducing banks' normal hedging costs from about 7%. **Special dollar swap window was opened for the PSU Oil firms** on Aug 28, 2013
- **Rebooking of cancelled forward exchange contracts** (for exporters and importers) clamped down.

Policy measures during Spillovers from Taper Talk Tantrum of 2013

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September 4, 2013 measures

- Limit of 25% for rebooking of cancelled forward exchange contracts for exporters was doubled to 50%; similar facility opened to importers with 25% limit
- A separate concessional swap window for attracting fresh FCNR(B) dollar funds of minimum 3-year tenor @3.5% was opened on September 4, 2013
- Overseas borrowings limit was hiked from 50% to 100% of unimpaired Tier I capital of the banks and a concessional swap facility (100 bps below prevailing swap rate) with the RBI for borrowings so mobilized
- India received US\$34.3 bn out of these two schemes (US\$26.0 bn in FCNR (B) and US\$8.3 bn in bank's overseas foreign currency borrowing). The swap window has helped rebuild buffer at a critical time after US\$11.6 bn was provided to public sector OMC through a swap facility with the RBI

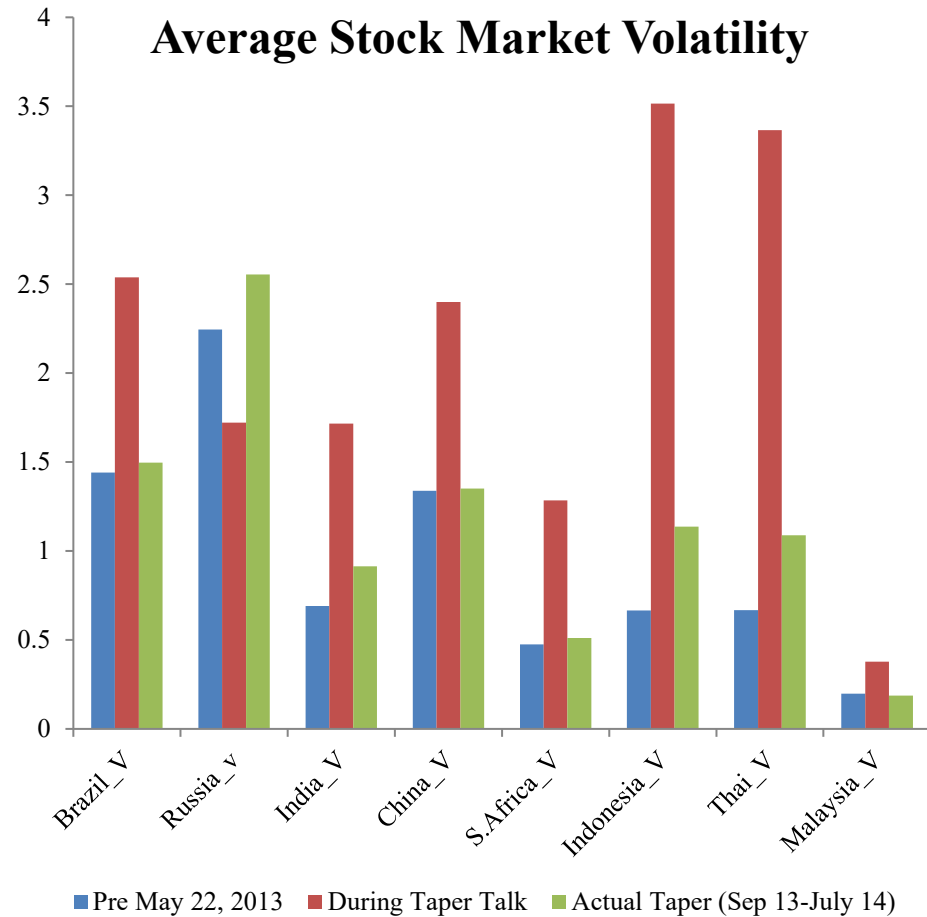
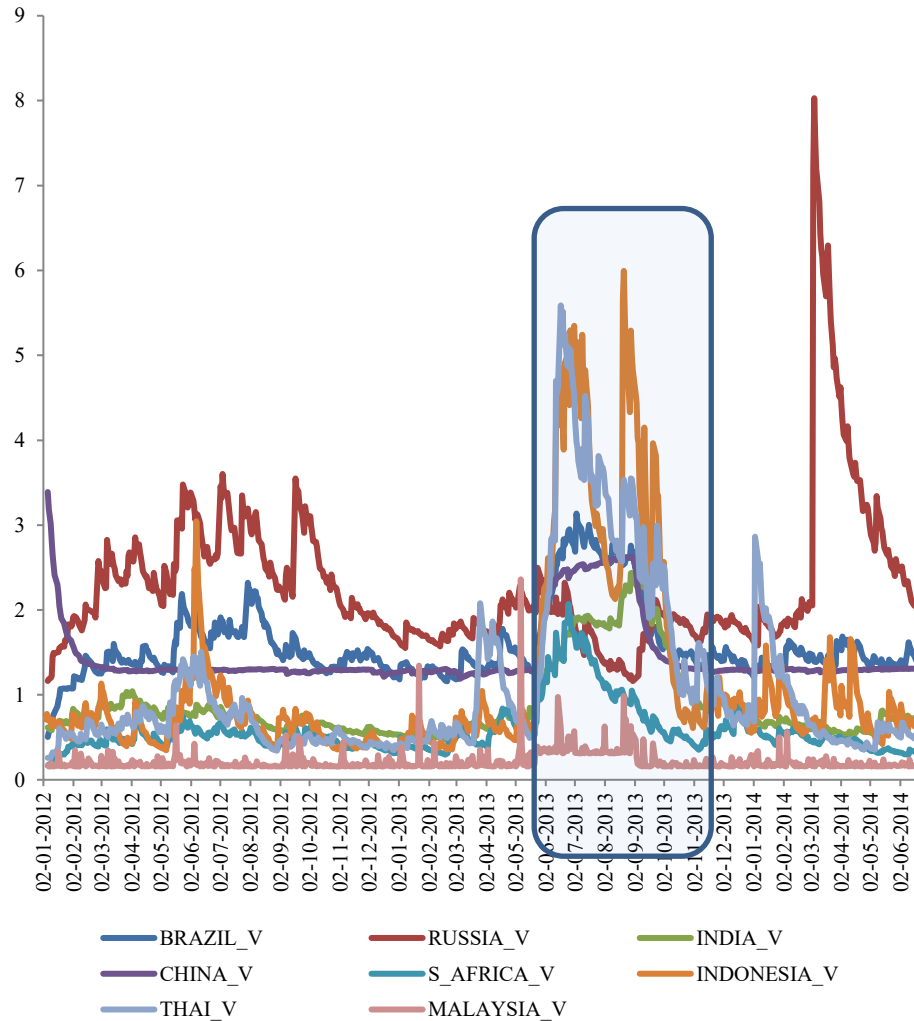
Augmented GARCH Model-Taper Tantrum Effects on India & Other EMEs

- $R_t = \alpha_0 + \alpha_1 * R_{t-1} + \varepsilon_t$ ε follow $N(0, H)$
- $H_t = \beta_0 + \beta_1 * \varepsilon_{t-1}^2 + \beta_2 * H_{t-1} + \delta * \text{Dum} + \zeta$
 - ▣ where ζ follows normal distribution
 - ▣ dummy (DUM) variable takes value one for *Fed's tapering talk*,
 - ▣ Dum-May 22, 2013 to August 2013 otherwise zero
 - ▣ estimated GARCH variance series (H_t) and the coefficient of the dummy variables (δ)
- Similar analysis also conducted through M-GARCH and

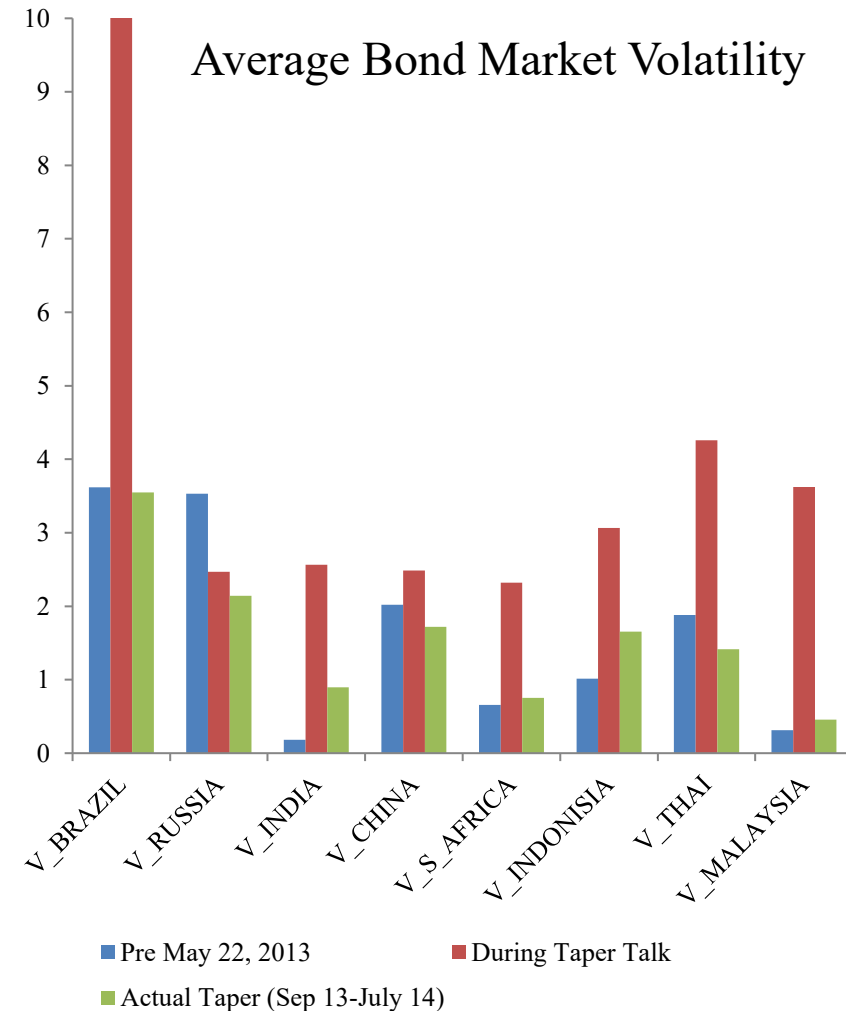
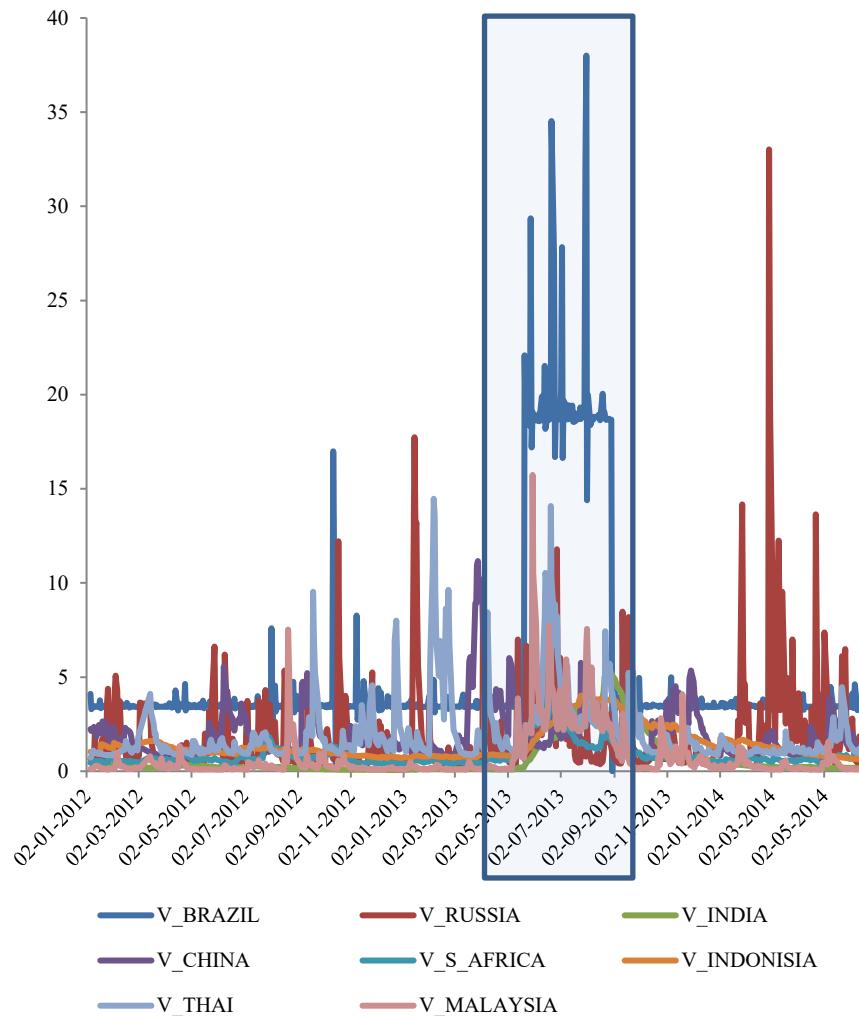
Augmented GARCH Model shows significant spillovers

- $R_t = \alpha_0 + \alpha_1 * R_{t-1} + \varepsilon_t$ ε follow $N(0, H)$
- $H_t = \beta_0 + \beta_1 * \varepsilon_{t-1}^2 + \beta_2 * H_{t-1} + \delta * \text{Dum} + \zeta$
 - ▣ where ζ follows normal distribution
 - ▣ dummy (DUM) variable takes value one for *Fed's tapering talk*,
 - ▣ Dum-May 22, 2013 to August 2013 otherwise zero
 - ▣ estimated GARCH variance series (H_t) and the coefficient of the dummy variables (δ)
- Similar analysis conducted through alternative multivariate GARCH models – MGARCH & BEKK; their results also find significant spillovers of volatility

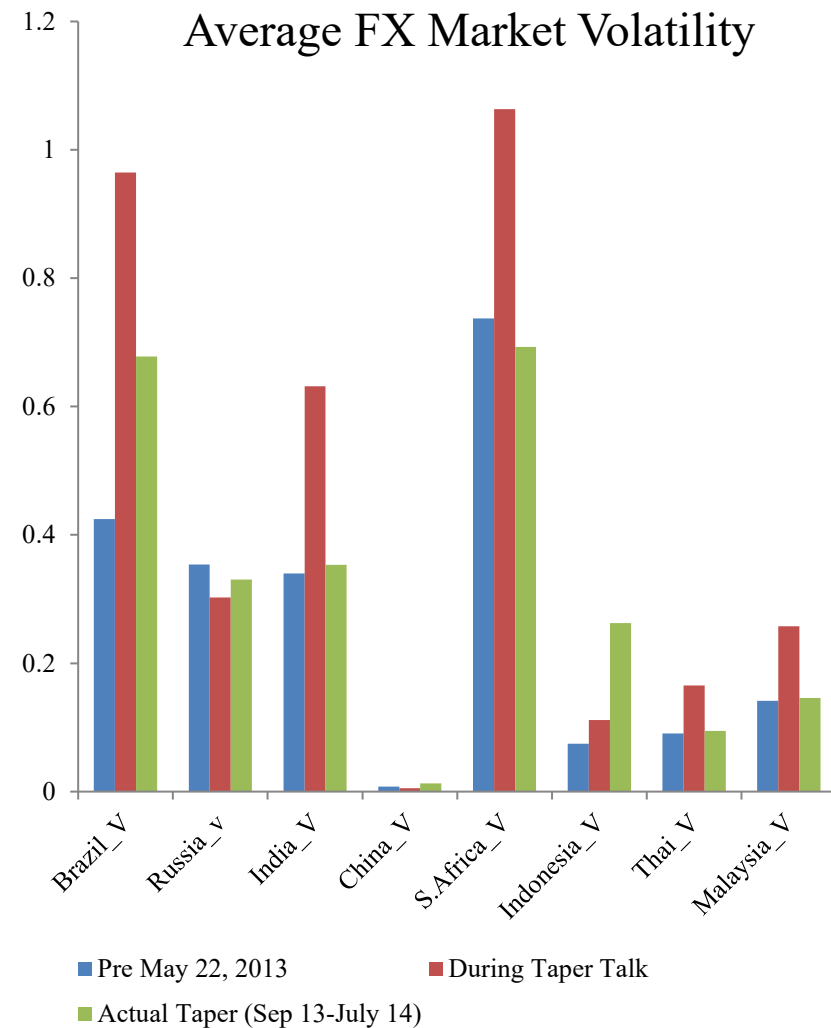
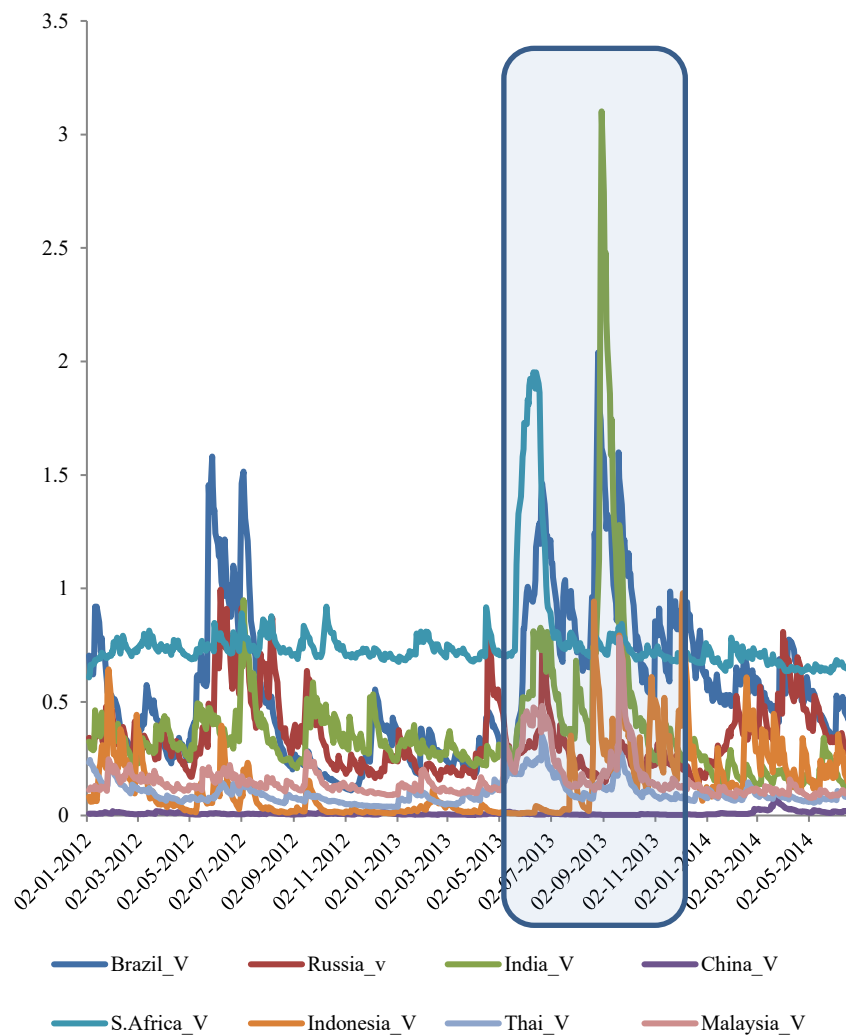
GARCH Conditional Variance in EME Stock Markets



GARCH Conditional Variance in EME Bond Markets



GARCH Conditional Variance in EME FX Markets



Lessons from the ‘Taper Tantrum’

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- Monetary policy communication is extremely important & needs to be handled with care in world where market sentiment shifts in a world of multiple equilibria. Clear and effective communication by AE central banks concerning exit from UMPs is important to reduce the risk of excessive volatilities. EM central bank is also important in imparting confidence to the market in a financial crisis-like situation.
- The Fed’s monetary policy announcements were strongly correlated with movements in asset prices and capital inflows in EMs, with the effects being largest during the phase of unconventional monetary policy (post-2008) and when tapering was first discussed (summer of 2013). Therefore, systemically important central banks need to consider spillovers and spillbacks of monetary policy.
- During initial periods of acute and systemic market volatility, asset prices and capital flows were hit indiscriminately across countries & investors behaved like a herd making no discrimination based on fundamentals. However, over time there was greater differentiation among EMs with good macroeconomic fundamentals helping in dampening market reactions to U.S. monetary policy shocks. In particular, elevated CAD, high inflation, weak growth prospects, and relatively low reserves were important factors affecting market reaction.
- Where vulnerabilities existed, EMs which acted early and decisively generally fared better.
- While to safeguard global financial stability, adequate GFSN with enough IMF resource power is necessary, macroeconomic stability and adequate FX reserves buffer is the best line of defence.